

Humanware & IT Governance

Lesson 47 — The Human Element and IT in Governance

JKSSB Computer Awareness

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What Is Humanware?

- ▶ A computer system needs people as much as it needs machines.
- ▶ **Humanware** (also called **liveware** or **peopleware**) is the human element of a computer system.
- ▶ It means the people who design, build, operate, manage, and use computers.
- ▶ Humanware is **neither hardware nor software**; it is the people component.
- ▶ Every other part of the system exists to serve the people who use it.

Definition

Humanware is the people involved with a computer system: its users, operators, administrators, programmers, and support staff.

The People in a Computer System

Role	What the person does
User / end user	Uses the computer to do day-to-day work
Operator	Runs the machine and routine jobs
Administrator	Manages systems, networks, databases, security
Programmer / developer	Writes and maintains software
IT support / technician	Installs, maintains, and troubleshoots

All of these people together are the humanware of the system.

Users, Operators, and Administrators

- ▶ **User (end user)**: the person who uses the finished system to perform a task — a clerk, a student, a customer.
- ▶ **Operator**: runs the computer system, feeds and monitors jobs, and handles routine operation.
- ▶ **System administrator**: manages servers, user accounts, and access rights.
- ▶ **Network / database administrator**: manages the network or the database.
- ▶ Administrators manage the system; operators run it; users use its output.

Programmers and IT Support

- ▶ **Programmer / developer**: writes, tests, and maintains the software the system runs.
- ▶ **IT support / help desk**: answers user problems and troubleshoots faults.
- ▶ **Technician**: installs, repairs, and maintains the hardware.
- ▶ All of these are humanware — the human element, not machinery.

One idea

Humanware is about people and their roles, not about devices or programs.

Hardware + Software + Humanware

Component	What it is
Hardware	The physical parts of the computer
Software	The programs and instructions
Humanware	The people who use and manage them

All three are required

Hardware without software does nothing, and both are useless without the people who run and use them.

IT in Governance: The Role

- ▶ **Governance** is the process of decision-making and its implementation in an organisation or a state.
- ▶ **ICT** (Information and Communication Technology) lets government deliver services over computers and networks.
- ▶ The **role of IT in governance** is to make government work faster, cheaper, more transparent, and more accessible.
- ▶ This is the “role of IT in governance” annexure topic.

Benefits of IT in Governance

- ▶ **Transparency**: citizens can see rules, status, and decisions.
- ▶ **Accountability**: actions are recorded and traceable.
- ▶ **Efficiency**: less paperwork and faster processing.
- ▶ **Convenience**: services available from home, round the clock.
- ▶ Lower cost of service delivery and reduced opportunities for corruption.

Takeaway

The aim is **citizen-centric** service — the citizen at the centre of every process.

IT Governance vs E-Governance (Near-Neighbour)

Term	Meaning
IT governance	Directing and controlling an organisation's own IT resources — governance <i>of</i> IT
E-governance	Using IT to govern and deliver public services — IT <i>in</i> governance

Trap alert

IT governance is **governance of IT**; e-governance is **IT in governance**. Do not swap the two.

E-Governance Basics: What It Is

- ▶ **E-governance** is the application of ICT by government to deliver services, share information, and transact with citizens, businesses, and other government agencies.
- ▶ It uses the internet, databases, and digital records instead of paper and counters.
- ▶ It makes government services faster, wider-reaching, and easier to audit.
- ▶ Named platforms and G2x matching are covered in detail in Lesson 57.

Definition

E-governance is the use of information and communication technology to improve the way government serves and interacts with the public.

G2C, G2G, G2B, G2E

Type	Full form	Interaction
G2C	Government to Citizen	Services to the general public
G2G	Government to Government	Between government departments
G2B	Government to Business	With companies and traders
G2E	Government to Employee	With government staff

These four are the standard interaction types of e-governance.

Digital Public Services

- ▶ Online delivery of government services: certificates, licences, land records, bill payment, applications, and grievance redressal.
- ▶ Citizens reach them through service portals and Common Service Centres (CSCs).
- ▶ Typical features: online application, status tracking, digital payment, and downloadable documents.
- ▶ A service is a “digital” public service when the citizen can complete the task without visiting an office in person.

One idea

Digital public services move a government task from a paper counter to an online process.

Electronic Records

- ▶ An **electronic record** is information created, stored, and retrieved in electronic (digital) form.
- ▶ Examples: e-documents, e-files, scanned papers, database entries, and email.
- ▶ Benefits: paperless storage, easy search and retrieval, sharing, and backup.
- ▶ An electronic record can have legal validity when it is authenticated, for example with a digital signature.

Authentication: Verifying Identity

- ▶ **Authentication** is verifying that a user, device, or system is who or what it claims to be.
- ▶ It answers the question: **“Who are you?”**
- ▶ Common methods: password, PIN, OTP, biometrics, and a digital certificate.
- ▶ Authentication must succeed before the system grants access.

Definition

Authentication is the process of confirming a claimed identity before access is allowed.

Authentication vs Authorization (Near-Neighbour)

Point	Meaning
Authentication	Proves identity — “who are you?”
Authorization	Grants or denies permission — “what may you do?”

Trap alert

Authentication **identifies**; authorization **permits**. Authentication comes first, then authorization.

Authentication Factors (2FA and MFA)

- ▶ **Something you know**: a password or PIN.
- ▶ **Something you have**: an OTP on your phone, a token, or a smart card.
- ▶ **Something you are**: a fingerprint, iris, or face.
- ▶ **Two-factor authentication (2FA)** uses two *different* factors; **multi-factor authentication (MFA)** uses two or more.
- ▶ Two passwords are not two factors — they are the same factor twice.

Trap Alert: Facts to Keep Straight

- ▶ Humanware = **people**; it is neither hardware nor software.
- ▶ IT governance = governance **of** IT; e-governance = IT **in** governance.
- ▶ G2C = Government to Citizen (not “Government to Company”).
- ▶ Authentication = identity; authorization = permission.
- ▶ An electronic record is digital information, not a printed paper file.

Lesson 47: Recall

- ▶ **Humanware** (liveware / peopleware) is the people element: users, operators, administrators, programmers, and IT support.
- ▶ Hardware, software, and humanware are all required for a working system.
- ▶ IT in governance improves transparency, accountability, efficiency, and citizen convenience.
- ▶ IT governance = governance of IT; e-governance = IT in governance; G2x = G2C, G2G, G2B, G2E.
- ▶ Digital public services, electronic records, and authentication are the building blocks of e-governance.
- ▶ Authentication proves identity; authorization grants permission.